Health hazard class	General description
Respiratory or skin sensitization	A respiratory sensitizer is a product that may cause allergy or asthma symptoms or breathing difficulties if inhaled. A skin sensitizer is a product that may cause an allergic skin reaction.
Germ cell mutagenicity	This hazard class includes products that may cause or are suspected of causing genetic defects. Genetic defects are permanent changes (mutations) to body cells that can be passed on to future generations.
Carcinogenicity	This hazard class includes products that cause or are suspected of causing cancer.
Reproductive toxicity	This hazard class includes products that may damage or are suspected of damaging fertility or the embryo, fetus, or offspring. Note: There is an additional category that includes products that may cause harm to breast-fed children.
Specific target organ toxicity – single exposure	This hazard class covers products that cause or may cause damage to organs (e.g., liver, kidneys, or blood) following a single exposure. This class also includes a category for products that cause respiratory irritation, drowsiness, or dizziness.
Specific target organ toxicity – repeated exposure	This hazard class covers products that cause or may cause damage to organs (e.g., liver, kidneys, or blood) following prolonged or repeated exposure.
Aspiration hazard	This hazard class is for products that may be fatal if they are swallowed and enter the airways.
Biohazardous infectious materials	These materials are micro-organisms (e.g., viruses, bacteria, or fungi), nucleic acids (e.g., DNA or RNA), or proteins that cause or are probable causes of infection, with or without toxicity, in humans or animals.
Health hazards not otherwise classified	This class covers products that are not included in any other health hazard class. These hazards occur following acute or repeated exposure and have adverse effects on the health of a person exposed to them — including injury or death. If a product is classified in this class, the hazard statement will describe the nature of the hazard.

Hazard categories

Each hazard class contains at least one category. The hazard categories are assigned a number (1, 2, etc.). Categories may also be called "types." Types are assigned an alphabetical letter (A, B, etc.). In a few cases, subcategories are also specified. Subcategories are identified with a number and a letter (for example, 1A and 1B).

Some hazard classes have only one category (for example, "Corrosive to metals"). Others may have two categories (for example, "Carcinogenicity" [cancer]) or three categories (for example, "Oxidizing liquids"). There are a few hazard classes with five or more categories (for example, "Organic peroxides").

Hazard category	Level of hazard
1	More hazardous
2A	
2B	
3	Less hazardous

The category tells you about how hazardous the product is (i.e., the severity of hazard).

- Category 1 is always the greatest level of hazard (i.e., it is the most hazardous within that class). If Category 1 is further divided, Category 1A within the same hazard class is a greater hazard than Category 1B.
- Category 2 within the same hazard class is more hazardous than Category 3, and so on.

There are a few exceptions to this rule. For example, for the "Gases under pressure" hazard class, the hazard categories are "Compressed gas," "Liquefied gas," "Refrigerated liquefied gas," and "Dissolved gas." These classes relate to the physical state of the gas when packaged and do not describe the degree of hazard.

In addition, the "Reproductive toxicity" hazard class has a separate category called "Effects on or via lactation." "Effects on or via lactation" was not assigned a specific numbered category. Reproductive toxicity also has categories 1 and 2, which relate to effects on fertility and/or or the embryo, fetus, or offspring. "Effects on or via lactation" is considered a different, but related hazard within the "Reproductive toxicity" class.

Preparing to classify a product

To prepare to classify a product, suppliers should:

(1) Refer to the classification criteria laid out in Part 2 of the Hazardous Products Regulations. (Follow the link to the HPR on page 93 of this book.)
(2) Identify the relevant hazard data for products.
(3) Review the data in light of the classification criteria to determine the appropriate hazard classes and categories. Note that specific guidance for classifying mixtures for health hazards is provided in Part 8 of the Hazardous Products Regulations.
(4) Document the rationale and information for future reference.

Classification must be determined based on comparison of all available hazard data to the criteria in the Hazardous Products Regulations. The data used must be scientifically sound and valid.

3 Pictograms

Overview of WHMIS 2015 pictograms

What's new

- Pictograms show the type of hazard at a glance.
- There are 10 pictograms.
- Most pictograms have a red, diamond-shaped border.
- Pictograms are assigned to specific hazard classes or categories.

Pictograms are graphic images that immediately show you what type of hazard a hazardous product presents. With a quick glance, you can see, for example, that the product is flammable, or if it might be a health hazard.

Most pictograms have a distinctive red, diamond-shaped border. Inside this border is a symbol that represents the potential hazard (fire, health hazard, corrosive, etc.). Together, the symbol and the border are referred to as a pictogram. Pictograms are assigned to specific hazard classes or categories.

The following table shows the pictograms. The name of each pictogram is in bold type. The words in the brackets describe the hazard.

Table 5. Introducing the pictograms



Exploding bomb
(for explosion or reactivity hazards)



Flame
(for fire hazards)



Flame over circle
(for oxidizing hazards)



Gas cylinder
(for gases under pressure)



Corrosion
(for corrosive damage to metals, as well as skin, eyes)



Skull and crossbones
(can cause death or toxicity with short exposure to small amounts)



Health hazard
(may cause or suspected of causing serious health effects)



Exclamation mark
(may cause less serious health effects or damage the ozone layer)



Environment*
(may cause damage to the aquatic environment)

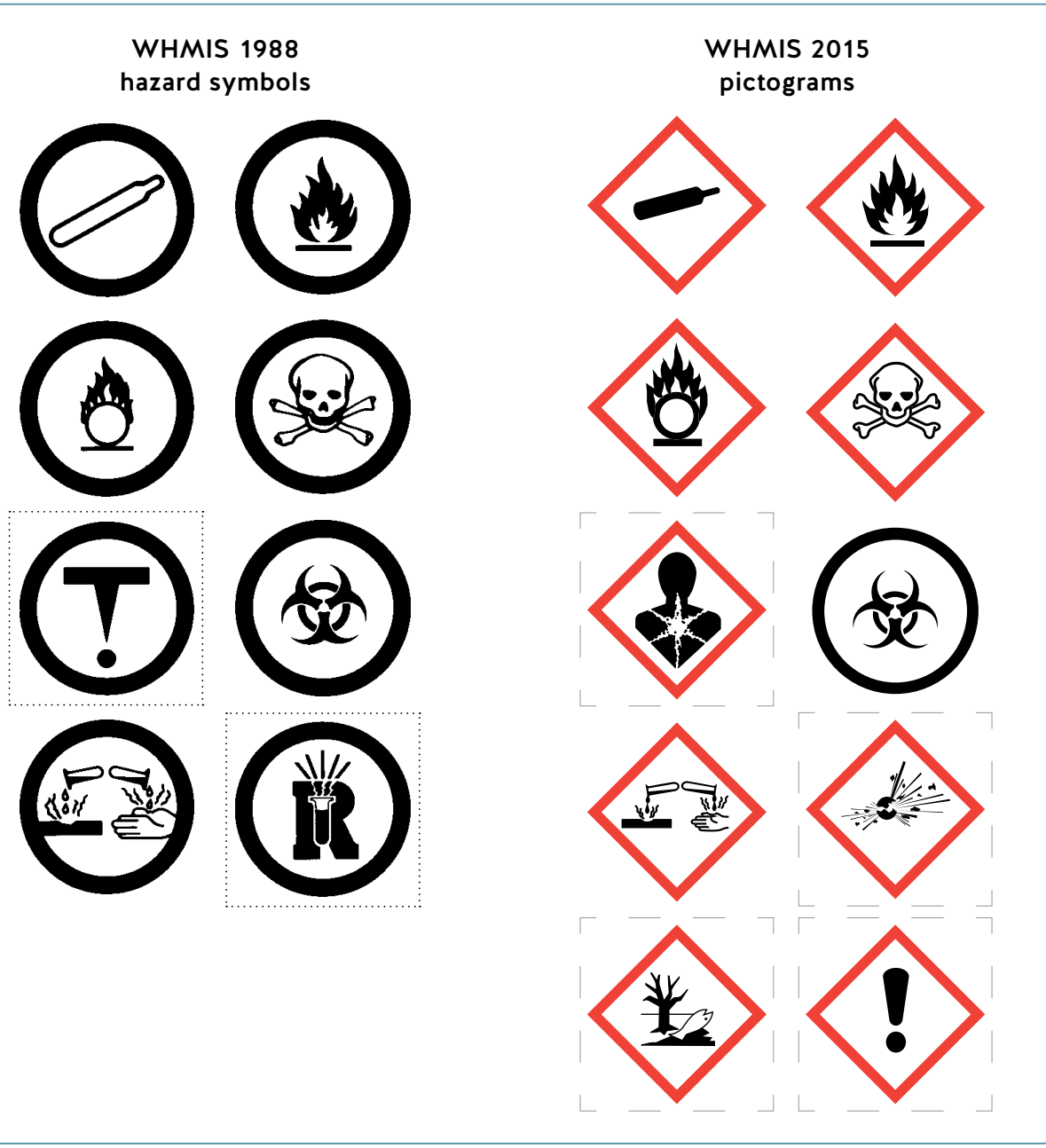


Biohazardous infectious materials**
(for organisms or toxins that can cause diseases in people or animals)

* An environmental hazards group exists in GHS. This group (and its classes) was not adopted in WHMIS 2015. However, you may see the environmental classes listed on labels and SDSs. Including information about environmental hazards is allowed by WHMIS 2015.

** The "Biohazardous infectious materials" hazard class was included in WHMIS 1988 but is not part of GHS. This class has been retained in WHMIS 2015 to maintain worker protection.

How pictograms compare to WHMIS 1988 hazard symbols



In general, pictograms (at right) are similar to WHMIS 1988 hazard symbols (at left). Many of the inner symbols are almost identical. However, there are some new symbols (the ones within the "Health hazard," "Exploding bomb," "Environment," and "Exclamation mark" pictograms, highlighted by dashed lines above right). And two symbols (in the "Materials causing other toxic effects" and "Dangerously reactive material" hazard symbols, highlighted by dotted lines above left) have been retired. In all but one case ("Biohazardous infectious materials"), the pictogram borders are red and diamond-shaped. And most pictograms are assigned to multiple hazard classes and categories.

How pictograms are used with WHMIS 2015 hazard classes and categories

For more information

For more information on the meaning of the categories, refer to Annex 1 (Classification and Labelling Summary Tables) of the United Nations publication *Globally Harmonized System of Classification and Labelling of Chemicals (GHS): Fifth revised edition.* unece.org/fileadmin/DAM/trans/danger/publi/ghs/ghs_rev05/English/05e_annex1.pdf

The pictograms are associated with the hazard classes and categories as follows.

Table 6. Pictograms matched to hazard classes and categories

Pictogram	Hazard classes and categories
	The **flame** pictogram is used for the following classes and categories: • Flammable gases (Category 1) • Flammable aerosols (Categories 1 and 2) • Flammable liquids (Categories 1, 2, and 3) • Flammable solids (Categories 1 and 2) • Pyrophoric liquids (Category 1) • Pyrophoric solids (Category 1) • Pyrophoric gases (Category 1) • Self-heating substances and mixtures (Categories 1 and 2) • Substances and mixtures which, in contact with water, emit flammable gases (Categories 1, 2, and 3) • Self-reactive substances and mixtures (Types B*, C, D, E, and F) • Organic peroxides (Types B*, C, D, E, and F)
	The **flame over circle** pictogram is used for the following classes and categories: • Oxidizing gases (Category 1) • Oxidizing liquids (Categories 1, 2, and 3) • Oxidizing solids (Categories 1, 2, and 3)

Pictogram	Hazard classes and categories
	The **gas cylinder** pictogram is used for the following classes and categories: • Gases under pressure (Compressed gas, Liquefied gas, Refrigerated liquefied gas, and Dissolved gas)
	The **corrosion** pictogram is used for the following classes and categories: • Corrosive to metals (Category 1) • Skin corrosion/irritation — Skin corrosion (Categories 1, 1A, 1B, and 1C) • Serious eye damage/eye irritation — Serious eye damage (Category 1)
	The **exploding bomb** pictogram is used for the following classes and categories: • Self-reactive substances and mixtures (Types A and B*) • Organic peroxides (Types A and B*)
	The **skull and crossbones** pictogram is used for the following classes and categories: • Acute toxicity: – Oral (Categories 1, 2, and 3) – Dermal (Categories 1, 2, and 3) – Inhalation (Categories 1, 2, and 3)
	The **health hazard** pictogram is used for the following classes and categories: • Respiratory or skin sensitization — Respiratory sensitizer (Categories 1, 1A, and 1B) • Germ cell mutagenicity (Categories 1, 1A, 1B, and 2) • Carcinogenicity (Categories 1, 1A, 1B, and 2) • Reproductive toxicity (Categories 1, 1A, 1B, and 2) • Specific target organ toxicity — Single exposure (Categories 1 and 2) • Specific target organ toxicity — Repeated exposure (Categories 1 and 2) • Aspiration hazard (Category 1)

Note

The "Physical hazards not otherwise classified" and "Health hazards not otherwise classified" classes are required to have a GHS pictogram that is appropriate to the hazard identified.

Pictogram	Hazard classes and categories
	The **exclamation mark** pictogram is used for the following classes and categories: • Acute toxicity — Oral, Dermal, Inhalation (Category 4) • Skin corrosion/irritation — Skin irritation (Category 2) • Serious eye damage/eye irritation — Eye irritation (Categories 2 and 2A) • Respiratory or skin sensitization — Skin sensitizer (Categories 1, 1A, and 1B) • Specific target organ toxicity — Single exposure (Category 3)
	The **biohazardous infectious materials** pictogram is used for the following classes and categories: • Biohazardous infectious materials (Category 1)

* Both the flame and exploding bomb pictograms are used for "Self-reactive substances and mixtures" (Type B) and "Organic peroxides" (Type B).

Hazard classes and categories without pictograms

Some hazardous products meet the criteria for hazard classes or categories but do not require pictograms. The product label and section 2 (Hazard identification) of the SDS still require the signal word, hazard statement(s), and other required label elements.

WHMIS 2015 hazard classes and categories that do not require a pictogram are:

- Flammable gases — Category 2
- Flammable liquids — Category 4
- Self-reactive substances and mixtures — Type G
- Organic peroxides — Type G
- Combustible dusts — Category 1
- Simple asphyxiants — Category 1
- Serious eye damage/eye irritation — Eye irritation — Category 2B
- Reproductive toxicity — Effects on or via lactation

Where to find pictograms

Pictograms will be on the supplier labels of the hazardous products you work with. They will also be on the SDSs (as the symbol or words that describe the symbol). For more information on labels, see Chapter 4 (starting on page 32). For more information on SDSs, see Chapter 5 (starting on page 48).

4 Labels

Labelling hazardous products

Under WHMIS 2015, hazardous products used, handled, or stored in the workplace must be labelled. Labels are the first alert to users about the major hazards of these products. They also outline the basic precautions or safety steps that should be taken.

In most cases, suppliers are responsible for labelling the hazardous products that they provide to customers. Employers are responsible for:

- Making sure that hazardous products that come into the workplace are labelled
- Preparing and applying workplace labels when appropriate

Overview of label types

There are two main types of WHMIS labels: supplier labels and workplace labels.

A **supplier label** is provided for each hazardous product by the supplier. Supplier labels will appear on all hazardous products received at a workplace in Canada. If a hazardous product is always used in its original container with a supplier label, no other label is required.

A **workplace label** is required when any of the following apply:

- A hazardous product is produced (made) at the workplace and used in that workplace
- A hazardous product is decanted (for example, transferred or poured) into another container
- A supplier label becomes lost or illegible (unreadable)

In general, employers are responsible for providing workplace labels. Employers must also ensure that all labels at their workplace are legible and that they are replaced if damaged.

Workers must be educated to recognize and understand the information provided on supplier labels, workplace labels, and other means of identification (such as warning signs, colour codes, and placards). Employers are responsible for developing this education and training in consultation with worker representatives or the occupational health and safety committee.

Supplier labels

What's new

- New requirements for supplier labels: pictograms, signal words, and standardized hazard statements and precautionary statements.
- Most hazard classes and categories have a prescribed pictogram, signal word, and hazard statement.

When supplemental label information is required

Supplemental label information is only included on a supplier label when:

- A toxic mixture has an ingredient with unknown acute toxicity, or
- A product reacts with water to produce an acutely toxic gas

Supplier labels continue to be required in English and French. They may be bilingual (as one label), or available as two labels (one in English, and one in French).

Information required

A supplier label must include the following information:

(1) **Product identifier** — the brand name, chemical name, common name, generic name, or trade name of the hazardous product.

(2) **Initial supplier identifier** — the name, address, and telephone number of either the Canadian manufacturer or the Canadian importer.*

(3) **Pictogram(s)** — hazard symbol within a red, diamond-shaped border.

(4) **Signal word** — a word used to alert the reader to a potential hazard and to indicate the severity of the hazard.

(5) **Hazard statement(s)** — standardized phrases that describe the nature of the hazard posed by a hazardous product.

(6) **Precautionary statement(s)** — standardized phrases that describe measures to be taken to minimize or prevent adverse effects resulting from:
   - Exposure to a hazardous product, or
   - Improper handling or storage of a hazardous product

In rare cases, supplier labels may also include **supplemental label information** that gives more details about precautionary actions, hazards not yet included in GHS, physical state, or route of exposure. This information must not contradict or detract from the standardized information.

* Initial supplier identifier — There are two exceptions to this requirement:
- In a situation where a hazardous product is being sold by a distributor, the distributor may replace the name, address, and telephone number of the initial supplier with their own contact information.
- In a situation where an importer imports a hazardous product for use in their own workplace in Canada (i.e., the importer is not selling the hazardous product), the importer may retain the name, address, and telephone number of the foreign supplier on the label and SDS instead of substituting their own contact information.

Format

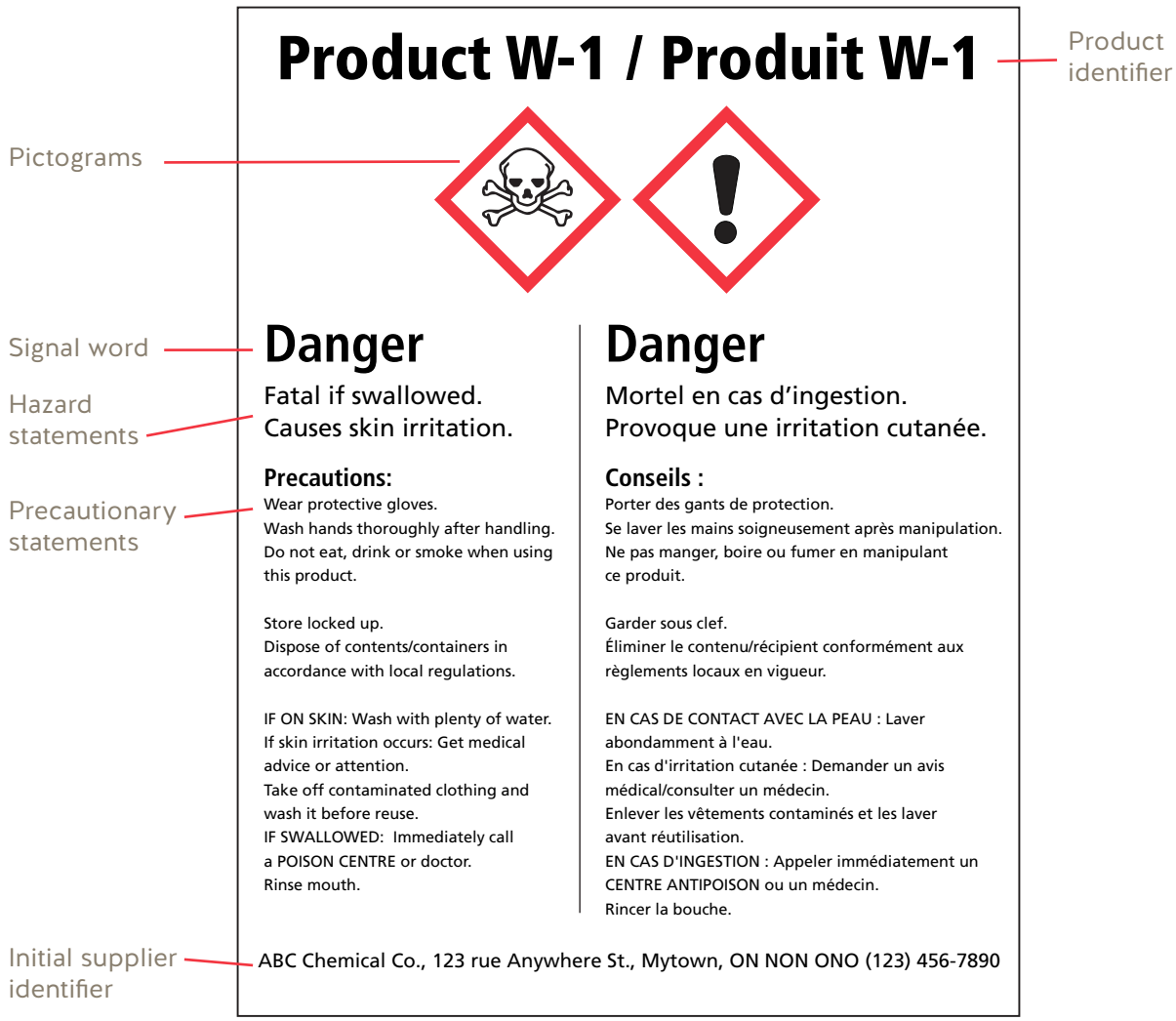
There is no set format for a supplier label, but the pictogram(s), signal word, and hazard statement(s) must be grouped together.

A supplier label must be:

- Clearly and prominently displayed on the container
- Easy to read (i.e., you can see it easily without using any item except corrective glasses)
- In contrast with other information on the product or container
- Bilingual (as one label or two)

An example of a bilingual label is shown below.

Sample supplier label



About signal words

A signal word is a prompt that alerts you about the degree or level of hazard of the product. There are only two signal words used: **Danger** or **Warning**. "Danger" is used for high-risk hazards, while "Warning" is used for less-severe hazards. If a signal word is assigned to a hazard class and category, it must be shown on the label and listed in section 2 (Hazard identification) of the safety data sheet (SDS).

Some hazard classes or categories do not have a signal word assigned to them.

About hazard statements

Each hazard class and category has an assigned "hazard statement." Hazard statements are brief, standardized sentences that tell you more about the exact hazard of the product. The statements are short, but they describe the most significant hazards of the product.

Examples of hazard statements are:

- Extremely flammable gas.
- Contains gas under pressure; may explode if heated.
- Fatal if inhaled.
- Causes eye irritation.
- May cause cancer.

The wording of the hazard statement helps to describe the degree of the hazard. For example, "May cause cancer" is more hazardous than "Suspected of causing cancer."

About precautionary statements

Precautionary statements provide advice on how to minimize or prevent adverse effects resulting from exposure to a hazardous product or resulting from improper storage or handling of a hazardous product. These statements can include instructions about storage, handling, first aid, personal protective equipment, and emergency measures. Like the hazard statements, the wording of precautionary statements is standardized.

For more information

For a list of all hazard classes and categories and the hazard statements assigned to them, refer to Annex 1 (Classification and Labelling Summary Tables) of the United Nations publication *Globally Harmonized System of Classification and Labelling of Chemicals (GHS): Fifth revised edition*. unece.org/fileadmin/DAM/trans/danger/publi/ghs/ghs_rev05/English/05e_annex1.pdf

There are four types of precautionary statements:

- Prevention
- Response (including first aid)
- Storage
- Disposal

Examples of precautionary statements are:

- Keep container tightly closed.
- Wear protective gloves/protective clothing/eye protection/face protection.
- If exposed or concerned: Get medical advice/attention.
- Fight fire remotely due to the risk of explosion.
- Protect from sunlight.

Precautionary statements will be consistent with the degree of the hazard associated with the product.

About slashes and dots on supplier labels

Slashes (/) and dots (...) are intended as instructions to the supplier to help them prepare the label and SDS.

For example, the guidance material from GHS lists the following precautionary statement: "Wear protective gloves/protective clothing/eye protection/face protection."

The slash (/) means the supplier is to specify the appropriate type of equipment based on their knowledge of the product and how it is used. So, for example, this statement could appear as:

- Wear protective gloves and eye protection.
  or
- Wear protective gloves.
  or
- Wear protective gloves, protective clothing, eye protection, and face protection.

"Do not subject to grinding/shock/friction/..." is another example. In this case, the supplier is to specify the applicable rough handling circumstance to avoid (grinding, shock, and/or friction), and the dots (...) mean they are to consider other types of rough handling that should be mentioned.

Updating supplier labels

Labels need to be updated when the supplier becomes aware of any "significant new data." Section 5.12 (1) of the Hazardous Products Regulations defines significant new data as:

> "...new data regarding the hazard presented by a hazardous product that change its classification in a category or subcategory of a hazard class, or result in its classification in another hazard class, or change the ways to protect against the hazard presented by the hazardous product."

Labels need to be updated within 180 days of the supplier being aware of the new information. If an employer purchases a product within this 180-day time period, the supplier must inform the employer of the changes, and the date they became available, in writing.

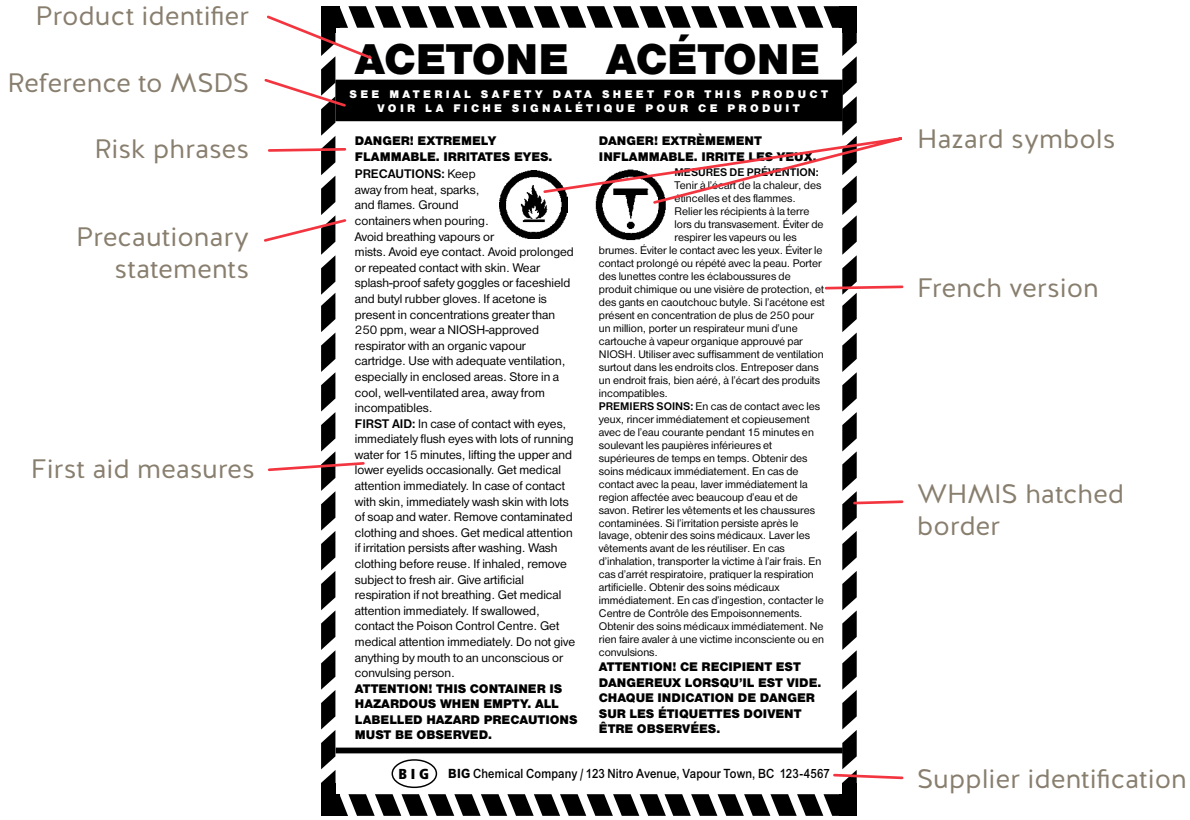
Employers need to update the existing labels or the information on the containers as soon as the supplier provides the significant new information.

Sample supplier labels: WHMIS 1988 vs. WHMIS 2015

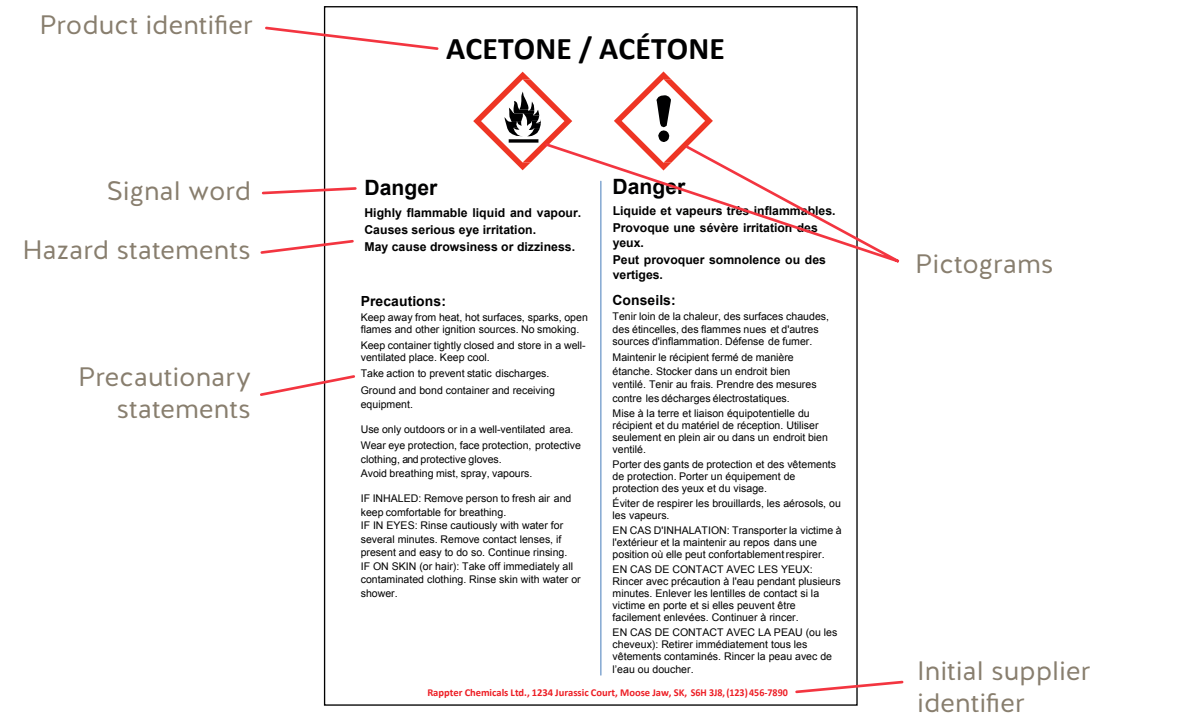
Two sample supplier labels are shown on the next page for comparison purposes: WHMIS 1988 (top) and WHMIS 2015 (bottom).

Comparing supplier labels

WHMIS 1988



WHMIS 2015



Supplier label requirements for shipments

When a supplier ships a product, it will generally be transported in a single container, in a multi-container, or in bulk. If the product is in a **single container**, the supplier must apply the supplier label. If a number of inner containers are packaged into a **multi-container shipment** (such as a box or wrapped pallet), the supplier must, in most cases, apply labels on both the inner and outer containers. For **bulk shipments** without packaging, the supplier is exempt from the labelling requirements of the Hazardous Products Regulations (i.e., supplier labels are not required).

The *Transportation of Dangerous Goods Act* may require additional labels during transport. For multi-container shipments, a supplier label is not required on the outer container if a Transportation of Dangerous Goods label is present. Only the inner containers require supplier labels.

Employers are responsible for checking that supplier labels have been applied to the hazardous products received at their workplaces. If the supplier sends labelling information instead of labels, the employer must develop and apply, at a minimum, a workplace label.

If a supplier label is missing when the product is received, or if the employer believes the label contains inaccurate information, the employer must temporarily store that product while he or she is actively seeking the information from the supplier or the manufacturer. The temporarily stored product cannot be handled or used until a proper label has been obtained.

If a supplier label becomes illegible or is accidentally removed, the employer must replace the label with either a supplier label or a workplace label.

Workplace labels

A workplace label provides the following required information:

- A product identifier identical to the one found on the hazardous product's safety data sheet
- Safe handling information for the hazardous product
- A reference to the availability of an SDS (if applicable)

Workplace labels may include pictograms or other supplier label information.

Format

The format for workplace labels is fairly flexible. For example:

- The information can be written directly onto the container using a permanent marker.
- The wording and language(s) used can be chosen to fit the specific workplace.

Sample workplace label

An example of a workplace label is shown below:

ACETONE

No smoking, sparks, or flames

Wear eye, face, and hand protection

Use in well-ventilated area, or wear NIOSH-approved respirator with organic vapour cartridges

Safety data sheet available

When workplace labels are required

As mentioned earlier, a **workplace label** is required when any of the following apply:

- A hazardous product is produced (made) at the workplace and used in that workplace
- A hazardous product is decanted (for example, transferred or poured) into another container
- A supplier label becomes lost or illegible (unreadable)

In general, employers are responsible for providing workplace labels. Employers must also ensure that all labels at their workplace are legible and that they are replaced if damaged.

Exceptions for decanted products
Workplace labels are not needed in two specific cases.

The first case is when a hazardous product is decanted from a container that has a supplier or workplace label on it into another container and:

- The decanted product stays under the control of the person who decanted it, and
- The decanted product's name (product identifier) is marked on the container, and
- All of the decanted product will be used during that same shift

For example, if a worker pours a hazardous product into a container (such as a jar, bottle, or bucket), writes the product's name on the container, will be the only person who will use the decanted product, and will use all of it during that same shift, then the worker doesn't need to apply a workplace label.

Note that an employer may have specific rules about labelling that go beyond what's required under WHMIS.

The second case is when a worker uses the decanted product immediately **and** completely. In this situation, the worker doesn't need to apply a workplace label or write the product's name on the container.

When employers produce hazardous products

If an employer *produces* a hazardous product at a workplace, the employer must ensure that a workplace label is applied to the hazardous product or its container. (Note: "Produces" does not include the escape of a hazardous product from equipment or from another product.) However, a workplace label is not required if the hazardous product is in a container that is intended to contain the hazardous product for sale or distribution and the container is or is about to be appropriately labelled.

Workplace label requirements fall under provincial or territorial jurisdiction, or under the *Canada Labour Code* in federally regulated workplaces.

Updating workplace labels

An employer must update a workplace label for a hazardous product as soon as significant new data about the product is available to the employer.